Rithish S

SDE Intern - Backend | Scalable APIs, Distributed Systems, Node.js & Databases

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Engineering & Campus Recruitment Team
CloudSEK Information Security Pvt. Ltd.
Bengaluru, Karnataka, India

Subject: Application for SDE Intern - Backend — Rithish S

Dear CloudSEK Engineering Team,

In an era where digital threats evolve at machine speed, CloudSEK's mission to build the world's fastest and most reliable AI technology for real-time threat analysis and remediation is nothing short of vital. From monitoring attack surfaces with BeVigil and protecting digital assets with XVigil to your pioneering launch of AIVigil—securing exposed AI infrastructure, vector databases, agentic workflows, and MCP servers—CloudSEK operates at the cutting edge of cybersecurity and artificial intelligence. As a backend-focused Computer Science undergraduate who builds resilient, secure distributed systems and contributes to world-class open-source tooling, I am eager to contribute to your backend systems from CloudSEK's Bengaluru office.

My core backend engineering experience aligns directly with CloudSEK's requirements for writing clean, efficient, and scalable code in Node.js. In **DB Backup CLI (DBVault)**—which I engineered and published to npm and Docker Hub with **120 automated tests**—I architected a multi-database backup utility supporting PostgreSQL, MySQL, MongoDB, and SQLite. To handle heavy I/O workloads predictably, I designed a distributed job-processing engine using **Redis and BullMQ** with configurable worker concurrency, queue rate limits, and Redis-based distributed locking. Because data integrity and confidentiality are paramount in cybersecurity, I implemented end-to-end **AES-256-GCM encryption**, local key rotation, and SHA-256 checksums before offloading to **AWS S3**.

To build scalable services that handle complex data flows, I architected **PatentIQ**, a high-performance backend microservice built with **Fastify**, **Prisma ORM**, and **PostgreSQL** adhering strictly to Clean Architecture and dependency injection. It powers a low-latency semantic search and RAG analysis pipeline integrating **Pinecone vector databases** and local LLMs (Qwen2.5 via Ollama)—experience that gives me practical empathy for the modern AI and vector infrastructure that CloudSEK's AIVigil actively monitors and protects. Furthermore, in my merged contribution to **ESLint Core (PR #21218)**, I resolved a subtle edge case in the `no-loss-of-precision` AST static analysis rule by distinguishing literal zeros from numbers that underflow under IEEE 754 binary64 specifications, writing regression suites that passed **43 automated CI verification checks**.

Complementing my backend and systems work is a foundation of rigorous problem-solving: I have solved **1200+ algorithmic problems** across LeetCode, CodeChef, and GeeksforGeeks, won 1st place in two collegiate hackathons (120+ and 200+ teams), and mentored **50+ students** in backend architecture and database modeling as a Web Developer for Students Special Group. I thrive in collaborative, fast-paced environments where code quality, peer review, and defensive engineering are celebrated.

I would welcome the opportunity to discuss how my backend engineering experience, cryptographic foundations, and curiosity can contribute to CloudSEK's backend products. Thank you for your time and consideration.

Sincerely,
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